INTRODUCTION

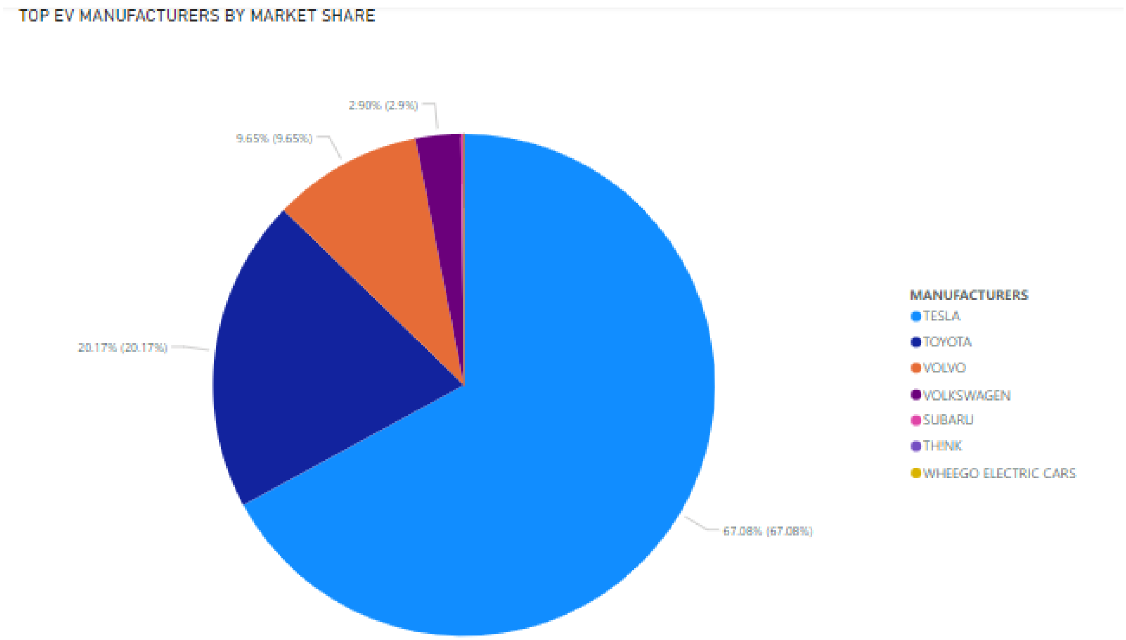
This report focuses on the analysis of electric vehicle (EV) adoption using a dataset sourced from Kaggle, which tracks the global population of electric vehicles. EVs, powered by electricity instead of traditional internal combustion engines, have gained popularity due to their environmental benefits, cost-effectiveness in the long term, and governmental support for reducing carbon emissions.

The dataset provides detailed information about the number of EVs across various countries, categorized by year, making it an invaluable tool for assessing the global shift toward electric mobility. To better understand these trends, I have utilized Microsoft Excel to create several visualizations that highlight the key patterns in EV adoption. The purpose of this report is to analyze the growth of electric vehicle popularity and explore the factors driving this change in transportation.

Dataset Source: Kaggle

Source link: https://www.kaggle.com/datasets/fatmanur12/electric-vehicle-population

Top EV Manufacturers by Market Share

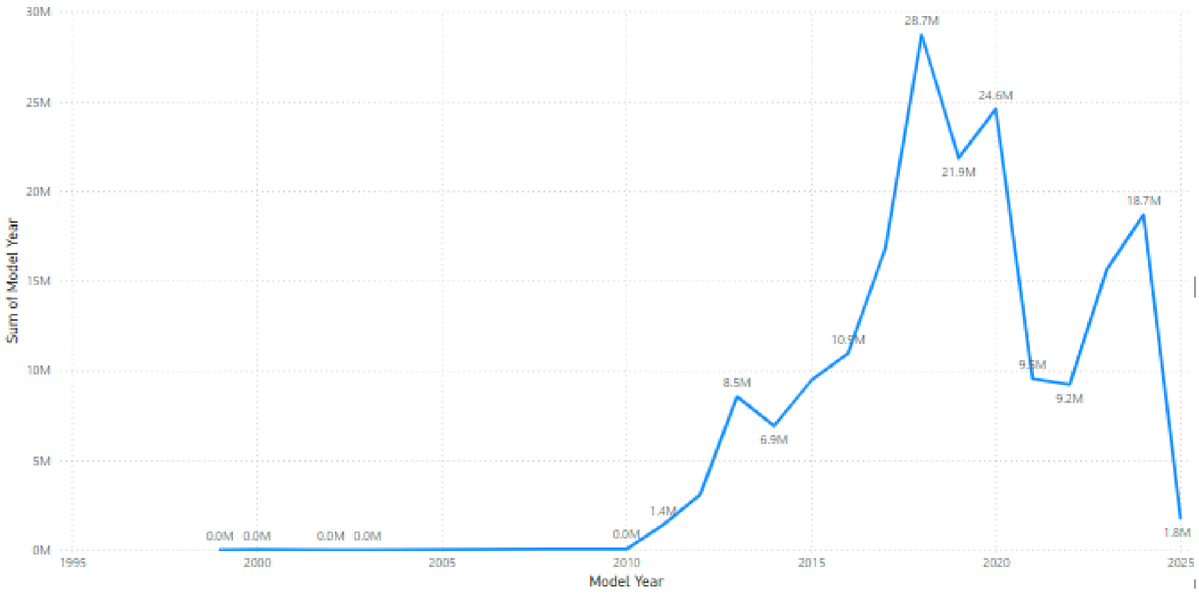


The pie chart, highlights the contributions of the **top 7 EV manufacturers** to the electric vehicle industry.

- **Tesla** leads the pack with a remarkable **67.08%** market share, cementing its status as the dominant player in the EV industry. Its stronghold reflects its pioneering technology, extensive global reach, and unmatched brand loyalty.
- **Toyota** holds the second spot with **20.17%**, showing its increasing focus on the EV segment. While significantly behind Tesla, Toyota's share signifies its potential to grow in this rapidly evolving market.
- **Volvo** comes in third, securing **9.65%** of the market. The company's dedication to sustainable and premium electric vehicles is evident, though it remains far behind the top two leaders.
- **Volkswagen** has a modest share of **2.90%**, indicating a growing but still developing presence in the EV space.
- The remaining manufacturers — **Subaru**, **Think**, and **Wheego Electric Cars** — hold minimal market shares. Their combined contributions showcase a more niche focus or limited influence in the broader market.

These top manufacturers show a steep disparity, with Tesla alone controlling more than two-thirds of the market. Toyota and Volvo, though sizable players, have considerable ground to cover. The smaller contributors, such as Subaru, Think, and Wheego, reflect the challenges faced by lesser-known or emerging manufacturers in carving out a meaningful market share.

Year-wise Growth of EV Registrations



Year-wise Growth of EV Registrations Analysis

The line chart titled illustrates the evolution of electric vehicle registrations over the years based on the sum of the model years. Here's a detailed breakdown:

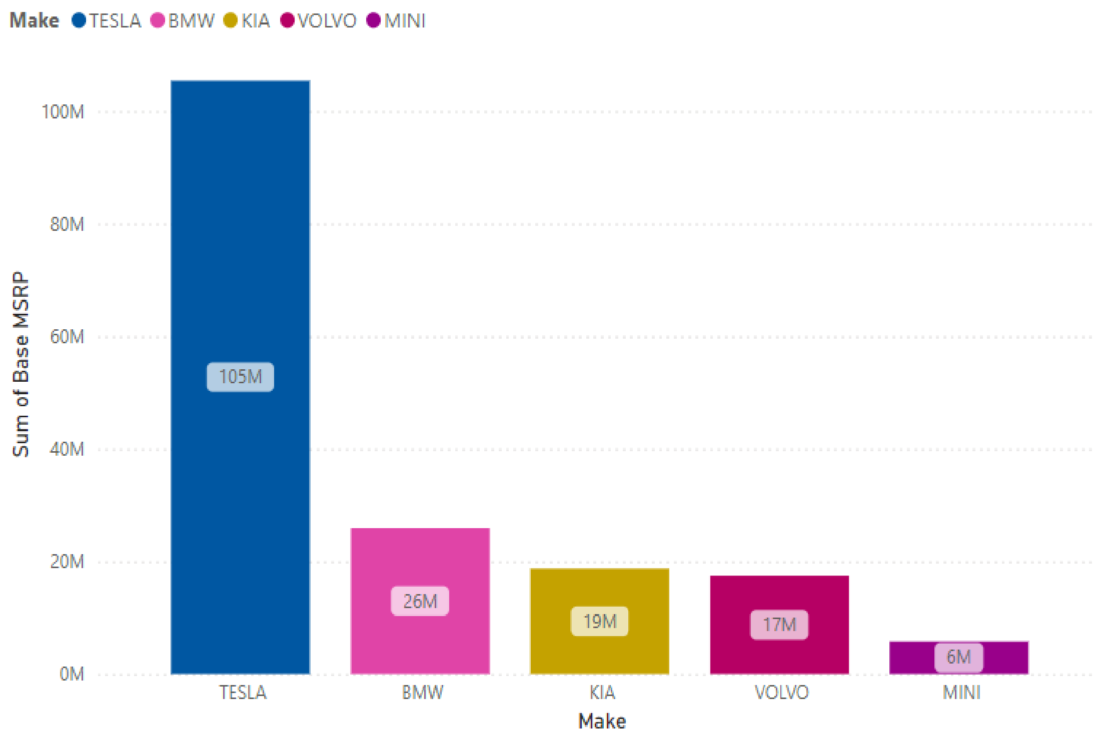
1. **Early Growth Phase (2005-2010)**:
   Registrations before 2005 are negligible. Around **2008**, we see the first signs of adoption, with **1.4 million registrations** recorded by 2010. This marks the initial phase of the EV market's growth.
2. **Steady Growth (2010-2015)**:
   The market gains momentum between **2010 and 2015**, reaching **10.4 million registrations**. This period highlights the growing acceptance of EVs, driven by improved technology, supportive government policies, and the introduction of more reliable EV models.
3. **Rapid Expansion (2015-2020)**:
   The EV market experiences a boom during this phase, peaking at **28.7 million registrations in 2018**. This rapid growth reflects significant global interest, advancements in EV technology, the expansion of charging infrastructure, and increased affordability.
4. **Fluctuations Post-Peak (2020-2025)**:
   After the 2018 peak, registrations dip to **9.2 million in 2020** before

recovering to **18.7 million in 2023**. A sharp drop to **1.8 million in 2025** is observed, but this is likely due to the fact that **2025 has only just started**, and data for the full year is incomplete.

- The period from **2015 to 2018** is the highlight of EV growth, showcasing the industry's success in scaling up.
- Post-2018 fluctuations could be attributed to supply chain challenges, external crises, or shifts in policy focus.

This analysis demonstrates the steady rise and the occasional challenges in the EV market, emphasizing its dynamic nature and the need for continuous investment and innovation.

Top 5 Car Manufacturers by Base MSRP in 2024



This chart visually compares the total base Manufacturer's Suggested Retail Price (MSRP) for the top 5 car manufacturers: Tesla, BMW, KIA, Volvo, and MINI. (Make suggests Manufacturers)

1. **Tesla Dominates in Price:** Tesla stands out with an exceptionally high "Sum of Base MSRP," surpassing the 100 million mark. This indicates that Tesla vehicles, on average, have significantly higher base prices compared to the

other manufacturers. This could be attributed to factors like advanced technology, premium features, and a strong brand image.

2. **BMW Follows at a Distance:** BMW occupies the second position with a "Sum of Base MSRP" around 26 million. While still commanding a premium price, it's notably lower than Tesla's. This suggests that BMW vehicles, while luxurious, might offer a slightly more accessible price point compared to Tesla.

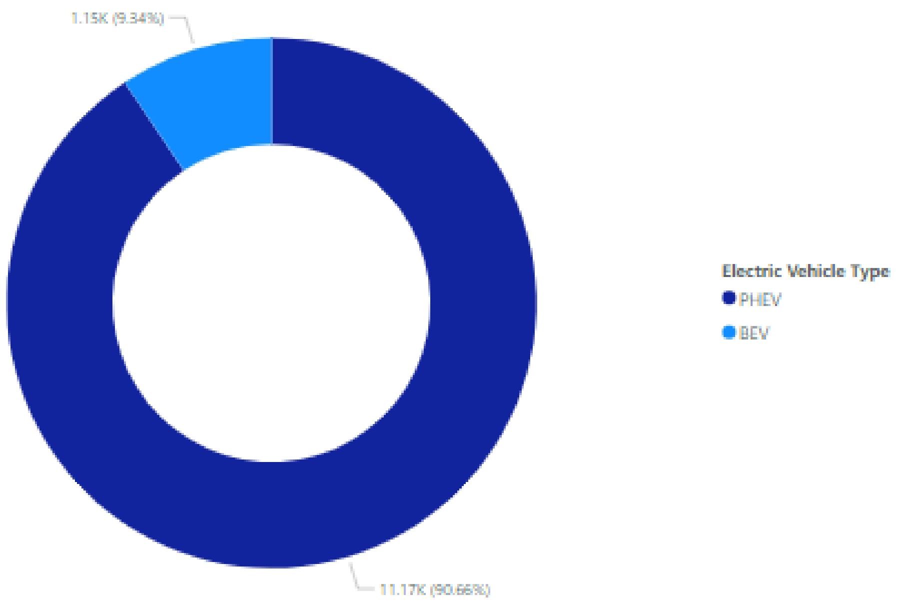
3. **KIA, Volvo, and MINI in the Mid-Range:** KIA, Volvo, and MINI fall into a lower price bracket. KIA holds the highest among these three, followed by Volvo and then MINI. This implies that these brands generally position themselves as offering more affordable options compared to Tesla and BMW.

Potential Implications:

- **Market Segmentation:** The price differences likely reflect the target audience for each brand. Tesla and BMW might primarily cater to high-income consumers seeking luxury and cutting-edge technology. KIA, Volvo, and MINI, on the other hand, might appeal to a broader range of buyers, including those looking for value-for-money options or specific features.

- **Brand Positioning and Value Proposition:** The higher MSRP for Tesla and BMW could be a strategic move to reinforce their premium brand image and position their vehicles as high-end luxury offerings. KIA, Volvo, and MINI, with their lower price points, might emphasize features like affordability, practicality, and specific niche appeal.

Distribution of EV Types in the Market



Plug-in Hybrid Electric Vehicles (PHEVs)

PHEVs combine a gasoline engine with an electric motor. This allows for both electric-only driving and the use of the gasoline engine for extended range. PHEVs offer flexibility and reduced emissions, making them a popular choice for many drivers. However, they generally have a higher initial cost and limited all-electric range compared to BEVs.

Market Share: PHEVs make up **90.66%** of the market, which signifies their overwhelming popularity and dominance in the electric vehicle space.

Battery Electric Vehicles (BEVs)

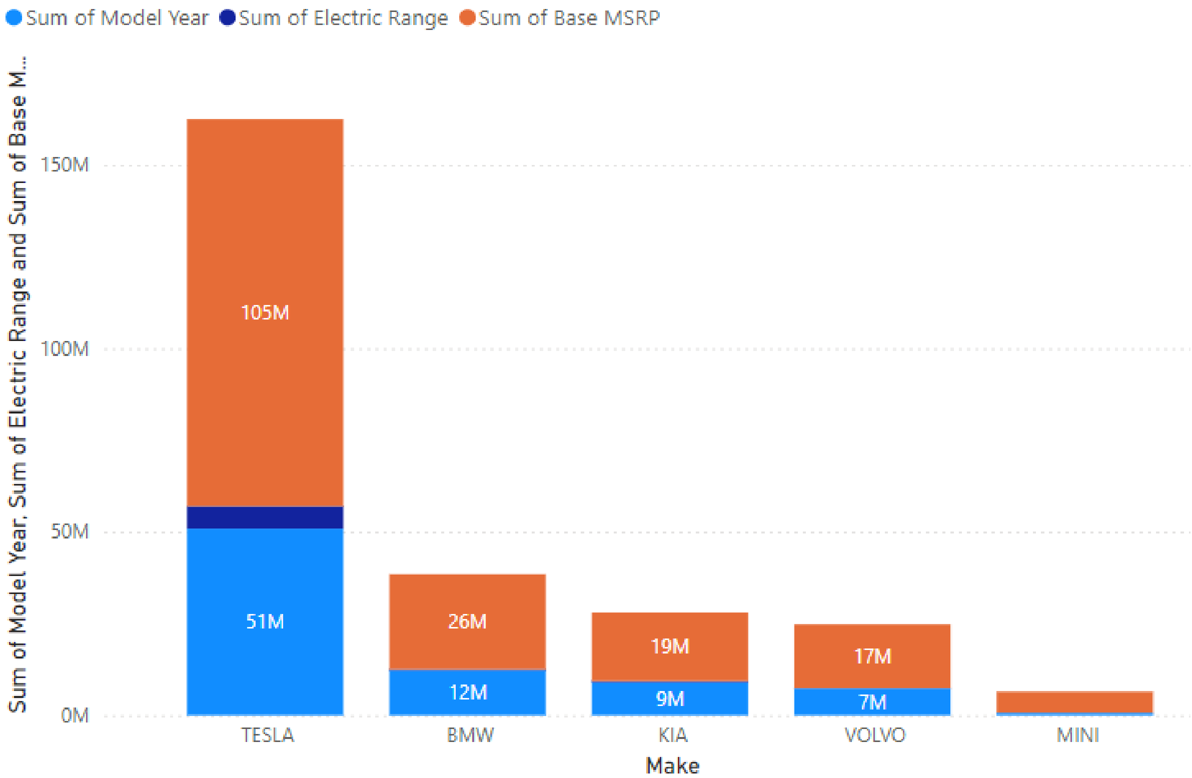
BEVs are fully electric vehicles that rely solely on a battery pack for propulsion. They offer zero tailpipe emissions, superior performance, and lower running costs. However, BEVs can have limitations such as range anxiety and the need for reliable charging infrastructure.

Market Share: Battery Electric Vehicles (BEVs) represent a smaller segment of the market, constituting only 9.34%.

In Summary:

PHEVs offer a balance of electric driving and traditional gasoline power, while BEVs provide a pure electric driving experience. The choice between the two depends on individual needs, driving habits, budget, and availability of charging infrastructure.

Comparison of Electric Range and Base MSRP by EV Manufacturer



This Chart compares the electric range and base Manufacturer Suggested Retail Price (MSRP) of electric vehicles (EVs) from leading manufacturers: Tesla, BMW, Kia, Volvo, and Mini

1. **Tesla** leads the market with a **51 million units electric range** and the highest **base MSRP of 105 million units**. Tesla's EVs offer exceptional range, making them ideal for long-distance travel. The premium pricing reflects advanced technology and superior performance.
2. **BMW** follows with an electric range of **26 million units** and a **base MSRP of 12 million units**. Offering a solid balance of range and affordability, BMW provides high-end features at a more accessible price point compared to Tesla.
3. **Kia** provides an electric range of **19 million units** and an affordable **base MSRP of 9 million units**. Kia offers competitive performance for those seeking an economical, reliable EV for daily use.
4. **Volvo** has an electric range of **17 million units** and a **base MSRP of 7 million units**. Offering a practical and safe option, Volvo positions itself

as a mid-tier choice for consumers seeking a balance of price and performance.

5. **Mini**, with the lowest range at **7 million units**, targets budget-conscious buyers with a **base MSRP of 2 million units**. Mini’s compact EVs are ideal for city driving, catering to consumers who need a short-range, affordable electric vehicle.

Tesla dominates the market in both electric range and price, targeting premium buyers. BMW offers a solid alternative, balancing range and price for those seeking luxury at a lower cost. Kia and Volvo provide affordable, reliable EVs with competitive ranges, while Mini focuses on providing entry-level EVs for city driving. As the EV market evolves, these manufacturers will continue to adapt to consumer needs, focusing on both performance and affordability.

Conclusion

In this report, I analyzed the electric vehicle (EV) population dataset to understand the trends and growth of EV adoption worldwide. The visualizations created in Excel have provided clear insights into the global EV market.

Top EV Manufacturers by Market Share (Pie Chart) showed the leading companies in the EV industry and their market shares.

Year-wise Growth of EV Registrations (Line Chart) highlighted the steady increase in EV registrations, reflecting growing adoption.

Top 5 Car Manufacturers by Base MSRP in 2024 (Bar Graph) compared the pricing of EVs from different manufacturers, showing the cost differences in the market.

Distribution of EV Types in the Market (Donut Chart) illustrated the popularity of various types of EVs, such as sedans and SUVs.

Comparison of Electric Range and Base MSRP by EV Manufacturer (Stacked Bar Chart) compared how the electric range and pricing of EVs differ by manufacturer.

Overall, the report shows a strong trend toward the adoption of electric vehicles, with manufacturers playing a key role in shaping the market. The increasing popularity of EVs, combined with a wider range of options at different price points, highlights the growing shift toward sustainable transportation.